

Ganga Singh Manchanda

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EDUCATION

University of Oxford DPhil, Quantum Algorithms, in collaboration with IBM	<i>Apr 2026 - Present</i>
Imperial College London MSc DIC, Quantum Fields and Fundamental Forces, Distinction	<i>Oct 2023 - Oct 2024</i>
Imperial College London BSc ARCS, Physics with Theoretical Physics, First Class Honours	<i>Oct 2020 - Jun 2023</i>

RESEARCH EXPERIENCE

University of Oxford <i>Doctoral Researcher — Physical and Theoretical Chemistry Laboratory</i>	Oxford, England <i>Apr 2026 - Present</i>
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IBM Research Europe <i>Doctoral Researcher — IBM Quantum</i>	Hursley, England <i>Apr 2026 - Present</i>
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Imperial College London <i>Graduate Researcher — Abdus Salam Centre for Theoretical Physics</i>	London, England <i>Oct 2024 - Mar 2025</i>
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- Co-authored a peer-reviewed paper published in *Physics Letters B* in the field of Canonical Quantum Gravity.
- Collaborated at every stage of the research process, from model building and theoretical analysis to interpretation of results, addressing key challenges along the way.

University of Birmingham <i>Student Researcher — Department of Mathematics</i>	Birmingham, England <i>Jul 2019 - Aug 2019</i>
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- Modelled thermoresponsive composite filaments using bead-spring simulations, to predict how micro-robotic shells can deform under localised heating.

PUBLICATIONS

- [1] G. S. Manchanda, “When energy isn’t enough”, *in prep. for Phys. Rev. A*.
- [2] G. S. Manchanda, “A note on complete gauge-fixing and the constraint algebra”, [[arXiv:2604.16990](https://arxiv.org/abs/2604.16990) [gr-qc]].
- [3] J. Magueijo and G. S. Manchanda, “Quantum wormholes at spatial infinity”, *Phys. Lett. B* **864**, 139434 (2025).

AWARDS

- EPSRC Industrial Doctoral Landscape Award — University of Oxford and IBM
- National Quantum Scholarship — Centre for Quantum Technologies, National University of Singapore (Declined)

TECHNICAL SKILLS

Programming Python (PennyLane, NumPy, SciPy, Pandas, Matplotlib, NetworkX), Git, Jupyter, L^AT_EX
Languages English, Punjabi, Hindi, French, Thai

ACTIVITIES & EXPERIENCE

4G Ventures Limited (Property Developer)

Project Manager

Birmingham, England

Jan 2025 - Present

- Managed the redevelopment of residential and commercial units, overseeing site operations and co-ordinating trades.
- Assessed acquisitions and advised on purchasing decisions for future projects.

Tutorful

Private Tutor

Online

Mar 2025 - Present

- Tutored students taking A-level and undergraduate examinations in Physics and Mathematics.
- Led students to improve their grades from the ranges B/C to A/A* (Enhanced DBS Certified).

Visely (AI EdTech Company)

Academic Writer

(Remote) London, England

Mar 2025 - Aug 2025

- Generated graduate-level exam questions in topics ranging from Riemannian Geometry to Quantum Information.
- Designed and refined LLM prompts to improve solution accuracy and academic rigour, ensuring high-quality content.

Imperial College Punjabi Society

President \ Vice-President

London, England

Oct 2022 - Jun 2024

- Oversaw operations and co-organized the UK's largest student cultural event (sold out Symphony Hall).
- Represented Imperial on stage in our annual inter-university dance competition (The Bhangra Showdown).

REFERENCES AVAILABLE UPON REQUEST